

Matias Barandiaran

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Education

Lancaster University

Bachelor of Science in Computer Science

Leipzig, Germany

October 2022 – July 2025

- Ranked **1st** out of **212** students.
- Final grade: **91.2%** (UK First Class; German grade **1.0**).
- University ranked **#132** worldwide for Computer Science (QS by subject) in 2025.

San Ignacio de Recalde School

International Baccalaureate

Lima, Peru

March 2020 – December 2022

- Cohort's top 5%.

Work Experience

Ransky Tech

Quantitative Developer

Remote

April 2026 – Present

- Working with quants and traders on research and strategy in proprietary crypto trading.

Technical University of Munich

Research Assistant (Hiwi)

Munich, Germany (Hybrid)

July 2024 – August 2025

- Hosted by the Gagneur Lab and Helmholtz Munich.
- Worked on rare variant association testing using deep learning and data-driven burden scores (DeepRVAT), utilizing **Python**, **R**, **PyTorch**, and **Snakemake**.

Martin Luther University of Halle-Wittenberg

Research Intern

Remote

July 2024 – September 2024

- Hosted by the Faculty of Medicine.
- Worked on graph-level anomaly detection via latent space searching and generative adversarial networks (gladGAN) on molecular datasets, utilizing **Python**, **PyTorch**, and **scikit-learn**.

Lancaster University Leipzig

Teaching Assistant

Leipzig, Germany

October 2023 – June 2024

- Delivered instructional sessions of **MIPS** assembly programming for the Digital Systems module.

Charles University

Research Intern

Prague, Czechia

August 2023 – September 2023

- Hosted by the Faculty of Mathematics and Physics.
- Contributed to paper “Unsatisfiability Proofs for Horn Solving”. *ETAPS best paper award*
- Designed production of Alethe verifiable proofs for the GOLEM horn clause solver in **C++**.

Freelance

Math Tutor

Lima, Peru

August 2020 – September 2021

- Guided senior high school students to achieve outstanding exam performance and substantial growth in their mathematical abilities (IB Mathematics SL and HL).

Publications

Barandiaran, M. & Stovold, J. (2025). Growing Reservoirs with Developmental Graph Cellular Automata. In Proceedings of International Conference on Artificial Life 2025 (pp. 26–35). MIT Press.

Otoni, R., Blichá, M., Barandiaran, M., Eugster, P., Kofrón, J., & Sharygina, N. (2025, May). Unsatisfiability Proofs for Horn Solving. In Proceedings of International Conference on Tools and Algorithms for the Construction and Analysis of Systems (pp. 67-87). Springer Nature.

Awards / Honours

Chancellor’s Medal – Lancaster University (2025): The university’s most prestigious award, presented annually to graduating students who demonstrate exceptional merit in their field.

Valedictorian – Lancaster University (2025): Ranked **1st** out of **212** students achieving the cohort’s highest average grade across all three years of the BSc Computer Science program.

Outstanding Dissertation – Lancaster University (2025): Obtained the highest achievable grade (**100%**) for the BSc Computer Science dissertation. **Strong Accept** for presentation at peer-reviewed International Conference on Artificial Life 2025 in Kyoto, Japan.

EASST Best Paper Award – Charles University (2025): Best paper related to the systematic and rigorous engineering of software and systems at peer-reviewed International Joint Conferences On Theory and Practice of Software 2025 in Hamilton, Canada.

Scholarship – Lancaster University (2022): Awarded a €4,000 tuition scholarship in recognition of academic merit.

Leadership Activities

Lancaster University	Leipzig, Germany
Founder, Computer Science Society	2023 – 2025

- Dedicated to fostering a culture of innovation, diversity and problem-solving.
- Secured university funding for the purchase of robotics supplies (e.g., 3D printer, Arduinos, etc.).

Lancaster University	Bailrigg, England
Co-Organizer, Global Advancement Fund	2024

- Co-Initiated and championed the approval process of a ~€20k fund to facilitate international collaboration between Lancaster campuses worldwide.

Lancaster University	Leipzig, Germany
Lead Organizer, Qiskit Fall Fest	2023

- Quantum Computing (QC) event powered by IBM.

- Gave an introductory talk on QC and held workshops with Qiskit material. Assembled a series of seminars with special guests like Enrique Solano (Co-CEO of Kipu Quantum).

Certifications

AWS Certified AI Practitioner – Amazon Web Services (2025): Demonstrated proficiency in applying foundational AI/ML concepts, model deployment workflows, and cloud-native AI services within AWS.

PyTorch for Deep Learning – DeepLearning.AI (2025): Hands-on training in building, fine-tuning, and deploying deep learning models in PyTorch across vision and NLP, covering modern architectures, optimization, interpretability, and production tools such as ONNX, MLflow, pruning, and quantization.

Prompt Engineering for Developers – DeepLearning.AI & OpenAI (2025): Covered techniques for designing effective prompts, controlling model behaviour, and building production-grade LLM applications through the OpenAI API.

Advanced Applications of Neural Networks and Deep Learning – University of Oxford (2023): Explored Generative Adversarial Networks, Variational Autoencoders, Deep Reinforcement Learning, Diffusion Models, and Graph Neural Networks.

Artificial Intelligence and Machine Learning: Theory and Practice – University of Oxford (2023): Studied the underlying mathematical foundations of linear and convolutional neural networks.

Additional Skills and Interests

Languages: Spanish (native), English (C2) *TOEFL: 114/120*, German (A2.1).

Tools: Python, Java, R, PyTorch, Lightning, TorchVision, Optuna, Git, C, C++, SQL, Linux, Snakemake, QEMU, Amazon Bedrock, Polars, smolagents, MLFlow.

Concepts: Agile Development, Continuous Integration, Data Pipelines, Computer Vision, Generative AI, Supervised Learning, Reinforcement Learning, Reservoir Computing, Cloud Computing, Retrieval-Augmented Generation, Natural Language Processing, Model Finetuning, Prompt Engineering, Agentic AI.

Hobbies: Game Development, Scuba Diving, Badminton.